Global-Integrated and Drift-Rectified Imprinting for Few-Shot Remote Sensing Object Detection

Bowei Yan, Gong Cheng, *Member, IEEE*, Chunbo Lang, Zhongling Huang, *Member, IEEE*, and Junwei Han, *Fellow, IEEE*

***Abstract*— Few-shot object detection (FSOD) in remote sensing images is a marginally explored but highly challenging task that focuses on identifying unseen classes of objects with a limited number of annotations. Current FSOD approaches often fail to accurately localize the foreground and misalign targets with various orientations, resulting in poor detection performance. For this purpose, we develop a fresh and powerful meta-learning framework based on the idea of imprinting, which leverages tailored support information to model the regional correlation between query and support objects in different stages. Specifically, a global-integrated scheme is first proposed to guide the generation of high-quality proposals by increasing the activation of foreground features and integrating global support information. Considering the orientation discrepancy of objects in query and support sets, we introduce a drift-rectified technique to achieve adaptive alignment by implicitly capturing the positional correspondence between the instances in two sets. In stark contrast to conventional FSOD approaches, our method can extract key clues and establish directional relationships between objects from different training sets, leading to better generalization capability. Extensive experiments on two standard benchmarks (DIOR and NWPU VHR-10.V2) manifest the effectiveness, and our proposed method exhibits superior performance to other competitors with similar motivation. The source code is available at https://github.com/Ybowei/GIDR**

***Index Terms*— Few-shot learning, few-shot object detection (FSOD), meta-learning, remote sensing images.**

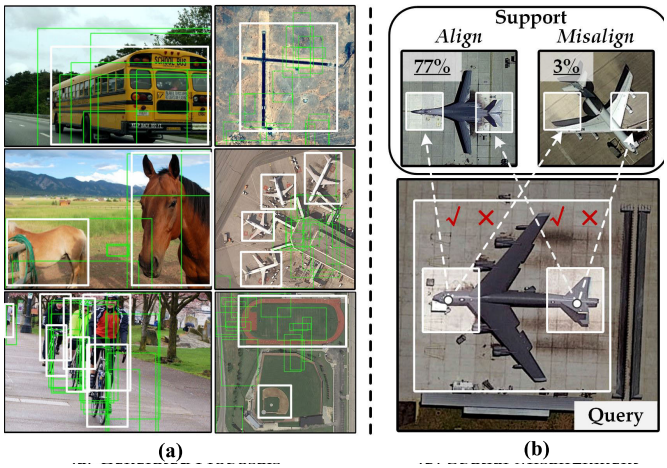


Fig. 1. (a) Visualization results of proposals on different kinds of images. Left: natural images. Right: remote sensing images. Unlike natural images, remote sensing images suffer from complex background noise, leading to low-scoring proposals in the foreground. The white and green boxes indicate the ground truth and generated proposals, respectively. (b) Spatial misalignment caused by the large variations in object orientation between the query and support sets. The matching score for airplanes positioned in the same direction is 77%, while for airplanes with different orientations is merely 3%.

I. Introduction

With the flourishing development of deep learning, significant progress has been made in remote sensing imagery interpretation, especially for scene classification [1], object detection [2], [3], [4], image segmentation [5], object tracking [6], and change detection tasks [7]. However, existing object detectors heavily rely on large-scale training samples to achieve desirable performance. Due to the lack of targets or the challenges in dataset annotation, training samples are often inadequate in many remote sensing contexts. Consequently, the object detectors learned with limited samples usually suffer from overfitting to training datasets and show unsatisfactory generalization performance.

To overcome this challenge, few-shot object detection (FSOD) has received increasing attention recently [8]. FSOD uses the knowledge acquired from base classes with sufficient annotations to enhance the performance of novel classes with limited data. However, the existing FSOD methods are mainly designed for natural images and are not directly applicable to remote sensing images. Compared with natural images, remote sensing images include complex backgrounds, and the targets show large directional variations [9]. These characteristics constantly hinder FSOD methods from accurately localizing the foreground and correctly matching the objects in the same category. For example, Fig. 1(a) indicates that the same FSOD method generates a markedly inferior quality of proposal in remote sensing images compared with its results in natural images. Furthermore, the matching scores of the query and support samples are significantly affected by the directional variations in remote sensing object targets, as illustrated in Fig. 1(b).

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Based on the observation, some prototype-based approaches are proposed to guide better generation of proposals for the foreground targets, such as Prototype-CNN [10] and MM-RCNN [11]. However, these approaches emphasize computational efficiency by encoding support images into global feature vectors, which leads to a loss of spatial information, thus compromising the quality of proposals. To address the mismatch between the targets of different branches, several attention-based methods [12], [13], [14], [15] are designed to align the query and support features using various attention mechanisms. Unfortunately, these methods have difficulty in modeling the positional relationship between the query and support objects, thus not being able to mitigate the category confusion caused by the diverse orientation of instances. Different from the above methods, there are certain FSOD approaches that apply straightforward image-level augmentation [10], [16] or feature-level augmentation [17] to reduce the discrepancy of orientation between query and support sets. Nevertheless, those handcrafted enhancements often suffer from limited diversity resulting in unsatisfactory detection performance.

In this article, we introduce an innovative meta-learning framework for FSOD, termed GIDR, which enhances foreground features by retaining the spatial information from the support set and aligns query–support features by establishing their positional and directional relationship from a local perspective. Our proposed GIDR greatly improves detection accuracy without relying on the manually designed augmentation strategy. Concretely, we propose two distinct modules for the tasks of different stages: global-integrated imprinting (GII) and drift-rectified imprinting (DRI). Building on the phenomenon that the query and support objects belonging to the same category share greater similarity than the background, the GII uses support features as convolutional kernels to activate the foreground features, thereby improving the quality of generated proposals of novel classes. Compared with previous attention-based approaches that aim to mine pixel-level correspondences, the GII focuses on the regional similarity between support and query features that mitigate the impact of background noise. Motivated by the insight that the objects within the same category typically consist of similar components, the DRI is designed to align the same part of the query and support objects through learnable offset compensation. In contrast to straightforward feature alignment approaches, the DRI is more adept at establishing the relative positional correspondences between the objects of two branches, and it also reduces the error delivery to the FSOD task. Our contributions can be summarized as follows.

1) We propose an innovative meta-learning framework for few-shot remote sensing object detection, which develops customized support information to exploit the correlation between query and support objects in different stages from a global-to-local perspective.
2) Considering the difficulty in identifying foreground targets in FSOD, we build a GII module to highlight foreground objects to provide better proposals for the detection head without any learnable extra parameters.
3) The DRI is designed to achieve adaptive alignment by implicitly capturing the positional correspondence between the query and support objects, thus increasing the matching accuracy.
4) Extensive experiments on the DIOR and NWPU VHR-10.v2 benchmarks show that our proposed method is superior to state-of-the-art models and significantly improves the robustness and generalization performance.

II. Related Work

A. Generic Object Detection

As a fundamental aspect of computer vision, object detection aims to accurately localize and identify objects in images [2], [3], [18], [19]. Meanwhile, it applies to a variety of downstream tasks, such as semantic segmentation [20], [21], [22] and vision-language comprehension [23], [24]. The existing object detection methods can be simply categorized into anchor-based [25], [26], [27] and anchor-free methods [28], [29], [30], depending on whether they use predefined anchors. Anchor-based methods decompose the task of object detection into two sequential stages. The initial step involves generating a series of region proposals likely to contain objects, after which classification and regression are performed on each proposal. In contrast, anchor-free methods directly predict bounding boxes and class labels through image features, eliminating the need for explicit region proposals. Despite recent promising results achieved by transformer-based detection models [31], [32] and large language models [33], [34], they are often constrained by high computing resource requirements and slow convergence time. Moreover, a common assumption across all these frameworks is sufficient annotated data from the known domain. This assumption leads to substantial challenges in the context of few-shot detection.

B. Few-Shot Object Detection in Natural Images

Recently, FSOD has attracted significant interest from researchers. Based on the training framework, current studies can be broadly divided into two main groups: transfer-learning-based approaches [35], [36], [37], [38] and meta-learning-based approaches [39], [40], [41], [42], [43], [44]. Transfer-learning-based methods typically pretrain a model with abundant annotations from base classes, followed by fine-tuning on novel classes with few labels. Although it provides a simple solution to addressing the challenges of FSOD partially, this straightforward training strategy frequently results in poor generalization capabilities. Meta-learning-based methods create meta-detectors to gain task-level insights by simulating few-shot situations with episodic dataset sampling. Although the training process is slightly complicated, it significantly improves the generalization ability of FSOD models. The essential principle of both the types of approaches is to take advantage of the knowledge gained from base classes to enhance detection performance for novel classes. However, due to the differences between remote sensing images and natural images, the detection methods designed for natural images cannot be directly adapted to the task of FSOD in remote sensing images.

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C. Few-Shot Object Detection in Remote Sensing Images

As with natural images, there has been an increasing focus on FSOD in remote sensing images [11], [13], [17], [45], [46], [47], [48], [49], [50] in recent years, reflecting the growing interest and advancements. Some prototype-based methods [10], [11] used support prototypes to provide better proposals for the detection head in FSOD models. However, encoding support images into global vectors leads to the loss of spatial information, thus affecting the quality of foreground proposals. Jiang et al. [45] developed an attention mechanism that establishes dependencies between the support and query, which inevitably introduces substantial noise, making it easier to confuse the foreground of novel classes with the background. Although FSOD has recently gained more attention, only a few existing approaches focus on the category confusion caused by the varied orientation of targets in remote sensing images. Cheng et al. [10] used a straightforward image-level augmentation technique that rotates each instance by five angles. Similarly, Zhang et al. [17] proposed a feature-level augmentation method that uses a designed dual-pipeline framework to produce category features. Nevertheless, those handcrafted enhancements often suffer from a limited diversity of generated samples, resulting in unsatisfactory detection performance. Other methods [13], [47], [48] introduced external knowledge to alleviate intraclass diversity, yet they failed to directly address the problem of variations in remote sensing object orientation. In contrast to these studies, our approach introduces two innovative interaction schemes between the query and support features, specially designed for the different tasks at each detection stage. It significantly increases the quality of foreground proposals and adaptively reduces the orientation discrepancy between query and support objects without relying on external knowledge, such as memory bank [48] or text modal knowledge [47]. Moreover, comprehensive experiments validate the effectiveness of each component and the generalization performance in our proposed method.

III. Methodology

A. Problem Setup

FSOD methods are expected to learn the transferable knowledge from training dataset D_{train} and exhibit good generalization on testing dataset D_{test} with just a small amount of labeled data. Formally, the training dataset $D_{\text{train}} = D_{\text{base}} \cup D_{\text{novel}}$ is divided into two parts: D_{base} with sufficient labeled data for the base classes C_{base} and D_{novel} with only a few annotated samples for novel classes C_{novel}. Current meta-learning-based methods typically optimize models through the episodic strategy. Specifically, each episode is made up of a support set $S = \{(\mathbf{x}_s^i, \mathbf{y}_s^i)\}_{i=1}^K$ and a query set $Q = \{(\mathbf{x}_q^i, \mathbf{y}_q^i)\}$, where $\mathbf{x}^*$ and $\mathbf{y}^*$ indicate a raw image and its corresponding annotation for a specific category c, respectively. In particular, when a support set includes N classes with K samples per category, this organization of training data is called N-way K-shot. Generally, the models are optimized through two training stages: base training and novel fine-tuning. Only the training samples from base classes are used during base training. In the novel fine-tuning stage, we randomly select an equal number of samples from both the base and novel classes to make up the fine-tuning set.

B. Method Overview

In this article, we introduce a novel meta-learning framework, termed GIDR, for the few-shot detection task. Unlike previous prototype-based and attention-based methods, our model focuses on extracting key clues from global to local levels through the idea of imprinting. Fundamentally, imprinting represents an adaptive learning mechanism driven by feature similarity, which uses relevant class information to define the relationship between the query and support objects [51], [52], [53]. Based on this concept, we develop a novel learning strategy, which leverages representative features from the support category to activate query features of the same class through convolution. Simultaneously, it significantly mitigates the impact of background noise in remote sensing images by modeling regional similarity via the designed interaction between the two branches. The general architecture of our method in the one-shot setting is illustrated in Fig. 2 without loss of generality. The proposed GIDR consists of three major components, i.e., a GII, a DRI, and a detection head. Specifically, the shared backbone of two branches is used to embed the query and support images, respectively. The fashioned GII module produces support kernels to interact with query features for the generation of foreground activation maps. Recognizing the substantial contribution that global support information provides to the quality of foreground proposal, the tailored ensemble module is used to refine the query features with the support prototypes by the guidance of foreground activation maps. This integration is then followed by obtaining the region of interest (RoI) features through the RPN and RoI-Align operation. Subsequently, we feed the RoI features to the DRI module, where several kernels and corresponding offsets are predicted to learn the correlation of spatial positions between the query and support objects. It performs adaptive feature alignment to improve matching accuracy without extra supervision. Finally, the enhanced RoI features are sent into the detection head to obtain the final detection results.

C. Global-Integrated Imprinting

Current FSOD methods typically use prototypes and attention mechanisms to guide the generation of foreground proposals. However, prototype-based methods [10], [11] generally perform max-pooling to obtain class prototypes, which ignores the critical spatial information of support sets for object localization. Attention-based methods [12], [45] mainly fuse support information by calculating pixelwise correlation. Unfortunately, these types of models inevitably introduce substantial background noise, leading to irrelevant feature activation. Based on this, we propose a GII module, which uses the different sizes of support kernels to highlight the key foreground features from a global view, as illustrated in Fig. 2.

1) Foreground Enhancement: Concretely, given a feature map $\mathbf{F}_s \in \mathbb{R}^{C \times H_s \times W_s}$ of support object, we first use an adaptive

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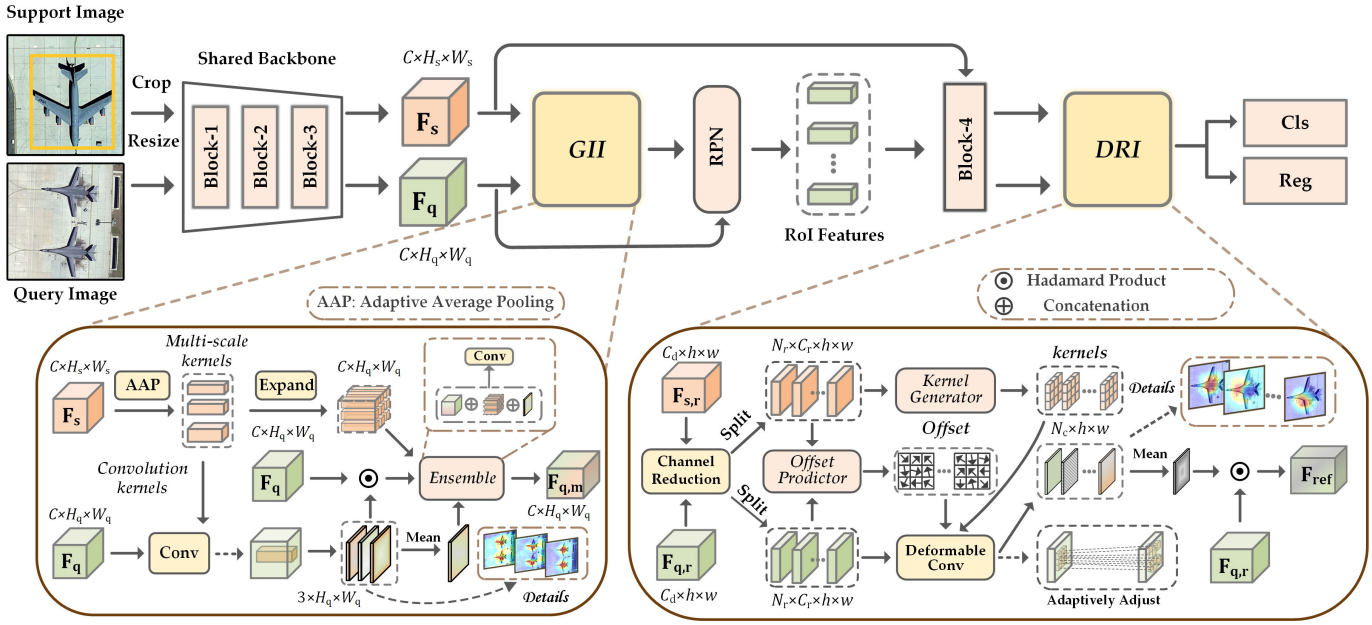


Fig. 2. Overall framework of our proposed GIDR for one-shot detection, including two primary contributions: GII and DRI. In each training episode, the shared backbone extracts the features of input image pair. The feature maps are then delivered to the GII module to strengthen the activation of foreground features, thereby facilitating better proposal generation. After acquiring the RoI features from query and support images, we input these features into the DRI module to achieve adaptive alignment of local components and use the detection head for class label and bounding box predictions.

pooling layer $\mathcal{F}_{\text{pool}}$ to generate support kernels $\mathbf{k}^{\theta}_{\text{fore}}$ with different scales, which is formulated as

$$\mathbf{k}^{\theta}_{\text{fore}} = \mathcal{F}_{\text{pool}}(\mathbf{F}_s) \in \mathbb{R}^{C\times m\times m}, \quad \theta \in 1, 2, \ldots, N_f \tag{1}$$

where m denotes the kernel size, and C is the number of channels in support feature $\mathbf{F}_s$. N_f indicates the number of kernels. Considering the varying object scales in remote sensing images, our work generates three different kernels with spatial sizes of 1×1, 3×3, and 7×7, respectively. Afterward, the activation maps are obtained by calculating the correlation between the query feature map $\mathbf{F}_q$ and different support kernels $\mathbf{k}^{\theta}_{\text{fore}}$ as follows:

$$\mathbf{A}^{\theta}_{\text{fore}} = \mathcal{F}_{\text{conv}}\left(\mathbf{k}^{\theta}_{\text{fore}}, \mathbf{F}_q\right) \in \mathbb{R}^{\left(\frac{H_q-m+2\times d}{t+1}\right)\times\left(\frac{W_q-m+2\times d}{t+1}\right)} \tag{2}$$

where $\mathcal{F}_{\text{conv}}$ means convolutional operation. d and t are the padding size and stride, respectively. To simplify calculations, the padding size d and stride t are assigned different values according to the kernel size m, ensuring that the generated activation map $\mathbf{A}^{\theta}_{\text{fore}}$ aligns with the spatial dimension of query feature $\mathbf{F}_q$. Then the final foreground activation map $\mathbf{A}_{\text{fore}}$ can be obtained by averaging and normalizing the different levels of activation maps

$$\mathbf{A}_{\text{fore}} = \text{softmax}\left(\frac{1}{N_k}\sum_{\theta=1}^{N_f}\mathbf{A}^{\theta}_{\text{fore}}\right) \in \mathbb{R}^{H_q\times W_q} \tag{3}$$

where N_f denotes the number of activation maps and the number of kernels with different sizes. The foreground activation map reflects the regional similarity between the query features and support object features to some extent. We use it to highlight the regions containing the targets, which can be described as

$$\mathbf{F}_{\text{q,e}} = \mathbf{F}_q \odot \mathbf{A}_{\text{fore}} + \mathbf{F}_q \tag{4}$$

where $\odot$ indicates the Hadamard product, and $\mathbf{F}_{\text{q,e}}$ represents the enhanced query feature. Essentially, the designed foreground enhancement strategy leverages the correlation filters to transform the target localization task into a global search problem. The contribution lies in finding better ways to evaluate the regional similarity between query and support image pairs to improve the detection performance without extra learnable parameters.

2) Information Concentration: To further enhance the response of the RPN to foreground objects, a feature ensemble module is applied to integrate the global support information under the guidance of activation maps $\mathbf{A}_{\text{fore}}$. For a given input feature $\mathbf{F}_s \in \mathbb{R}^{C\times H_s\times W_s}$, the expanded global support feature can be obtained

$$\mathbf{F}_{\text{s,g}} = \zeta_{H\times W}\left(\mathcal{F}_{\text{avgpool}}(\mathbf{F}_s)\right) \in \mathbb{R}^{C\times H_q\times W_q} \tag{5}$$

where $\zeta_{H\times W}(\cdot)$ expands the given vector to a spatial size $H \times W$ and $\mathcal{F}_{\text{avgpool}}$ is the global-average pooling operation. After that, we concatenate the enhanced query feature $\mathbf{F}_{\text{q,e}}$, expanded global support feature $\mathbf{F}_{\text{s,g}}$, and the foreground activation map $\mathbf{A}_{\text{fore}}$, and then the merged query features $\mathbf{F}_{\text{q,m}}$ are derived from a convolutional layer as follows:

$$\mathbf{F}_{\text{q,m}} = \mathcal{F}_{1\times 1}\left(\mathbf{F}_{\text{q,e}} \bigoplus \mathbf{F}_{\text{s,g}} \bigoplus \mathbf{A}_{\text{fore}}\right) \tag{6}$$

where $\mathcal{F}_{1\times 1}$ denotes a 1×1 convolutional layer that integrates the support and query information, and $\bigoplus$ represents the concatenation operation along channel dimension.

D. Drift-Rectified Imprinting (DRI)

As discussed in Section II-C, previous methods adopted image-level [10] or feature-level [17] augmentation to alleviate the mismatch between query and support targets caused by

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orientation diversity. However, these technologies commonly exhibit restricted diversity, resulting in unsatisfactory detection performance. Unlike the above approaches, we present a novel and effective module called DRI, which is specifically designed to reduce the directional discrepancies by capturing the correlations between the components of query and support objects from a local perspective.

As shown in Fig. 2, the lightweight DRI mainly consists of a kernel generator, an offset predictor, and an adaptive alignment module. After producing the query RoI features $\mathbf{F}_{\text{q,roi}}$ and support RoI features $\mathbf{F}_{\text{s,roi}}$ by the RoI-Align operation, a 1×1 convolutional layer is first applied to reduce the channel dimension for computational efficiency and generate the intermediate feature maps

$$\begin{cases} \mathbf{F}_{\text{q,r}} = \mathcal{F}_{1\times1}(\mathbf{F}_{\text{q,roi}}) \in \mathbb{R}^{C_d \times h_q \times w_q} \\ \mathbf{F}_{\text{s,r}} = \mathcal{F}_{1\times1}(\mathbf{F}_{\text{s,roi}}) \in \mathbb{R}^{C_d \times h_s \times w_s} \end{cases} \tag{7}$$

where C_d is the reduced channel. We then group $\mathbf{F}_{\text{q,r}}$ and $\mathbf{F}_{\text{s,r}}$ along the channel dimension to effectively mitigate the overfitting issue commonly encountered in FSOD by reducing the number of parameters, which can be formulated as

$$\begin{cases} \mathbf{F}_{\text{q,r}} = \left[\mathbf{F}^1_{\text{q,r}}, \mathbf{F}^2_{\text{q,r}}, \ldots, \mathbf{F}^{N_r}_{\text{q,r}}\right] \in \mathbb{R}^{(C_r \times N_r)\times h_q \times w_q} \\ \mathbf{F}_{\text{s,r}} = \left[\mathbf{F}^1_{\text{s,r}}, \mathbf{F}^2_{\text{s,r}}, \ldots, \mathbf{F}^{N_r}_{\text{s,r}}\right] \in \mathbb{R}^{(C_r \times N_r)\times h_s \times w_s} \end{cases} \tag{8}$$

where N_r is the number of groups, $\mathbf{F}^i_{*,\text{r}}$ denotes the feature map of the ith group, and C_r is the number of channel in $\mathbf{F}^i_{*,\text{r}}$. Unlike the scheme proposed in GII, which directly uses adaptive pooling to obtain support kernels, we introduce a kernel generator $\mathcal{F}_{\text{kernel}}$ that chooses different support information to learn relevant kernels from the local level

$$\mathbf{k}^i_{\text{roi}} = \mathcal{F}_{\text{kernel}}\left(\mathbf{F}^i_{\text{s,r}}\right) = \mathcal{F}_{1\times1}\left(\mathcal{F}_{3\times3}\left(\mathbf{F}^i_{\text{s,r}}\right)\right) \tag{9}$$

where $\mathcal{F}_{3\times3}$ is a 3×3 convolutional layers with C_r kernels. In our article, we select the 3×3 kernel size to optimize the detection results.

Inspired by [54], we apply a deformable convolution to achieve adaptive alignment for category match. In contrast to the default scheme that obtains the offsets of kernels from their original image features, the offsets ϕ_r learned from our designed predictor $\mathcal{F}_{\text{offset}}$ are mapped from the feature correction between the query and the support objects, which can be summarized as follows:

$$\phi^i_r = \mathcal{F}_{\text{offset}}\left(\mathbf{F}^i_{\text{q,r}}, \mathbf{F}^i_{\text{s,r}}\right) = \mathcal{F}_{3\times3}\left(\mathbf{F}^i_{\text{q,r}} \bigoplus \mathbf{F}^i_{\text{s,r}}\right) \tag{10}$$

$$\phi_r = \left[\phi^1_r, \phi^2_r, \ldots, \phi^i_r, \ldots, \phi^{N_r}_r\right]. \tag{11}$$

The offset ϕ_r represents the spatial difference between $\mathbf{F}_{\text{q,roi}}$ and $\mathbf{F}_{\text{s,roi}}$. Similar to the acquisition of $\mathbf{A}_{\text{fore}}$, the learned kernels and offsets are fed into the deformable convolution layer $\mathcal{F}_{\text{deform}}$ to produce an aligned activation map

$$\mathbf{M}_{\text{align}} = \text{softmax}\left(\frac{1}{N_r}\sum_{i=1}^{N_r}\mathcal{F}_{\text{deform}}\left(\mathbf{k}^i_{\text{roi}}, \mathbf{F}^i_{\text{q,r}}\right)\left[\mathbf{p} + \phi^i_r\right]\right) \tag{12}$$

where $\mathbf{p}$ denotes the basic sampling positions corresponding to the offsets. Applying (4) to $\mathbf{M}_{\text{align}}$ and $\mathbf{F}_{\text{q,roi}}$, we finally get the refined RoI feature map $\mathbf{F}_{\text{q,refine}}$. Importantly, the alignment achieved by DRI does not seek a rigid feature match like previous approaches [55], [56], instead it adaptively captures the key regional clues to match the same class between the query and support objects.

E. K-Shot Extension

To improve computational efficiency, our proposed method mainly addresses the K-shot setting by averaging activation maps. Specifically, for each support feature, the foreground activation map $\mathbf{A}_{\text{fore}}$ is generated using (3), while the aligned activation map $\mathbf{M}_{\text{align}}$ is acquired through (12). By computing the averaged values of two distinct types of activation maps, we obtain the final foreground activation map $\mathbf{A}_{\text{avg}}$ and the aligned activation map $\mathbf{M}_{\text{avg}}$. In addition, in the ensemble module of GII, the support prototype is expanded to integrate with the averaged activation map and the enhanced query feature through concatenation.

IV. Experiments

A. Datasets and Evaluation

1) Datasets: The proposed method is evaluated on two standard FSOD benchmarks, i.e., DIOR [60] and NWPU VHR-10.v2 [61]. The former is originally designed for the remote sensing object detection task, which consists of 20 categories: Baseball field (BF), Basketball court (BC), Bridge (BR), chimney (CH), Ship (SH), Airplane (APL), Airport (APO), Expressway toll station (ETS), Harbor (HA), Ground track field (GTF), Dam (DAM), Golf field (GF), Storage tank (STO), Tennis court (TC), Vehicle (VE), Expressway service area (ESA), Overpass (OP), Stadium (STA), Train station (TS), and Windmill (WM). Those classes are divided equally into four splits: $\{\text{DIOR-20}^i, i \in \{0, 1, 2, 3\}\}$. The specific settings for base and novel classes in each split follow [10], [11], [46], [47], [48], as illustrated in Table I. The latter, built from the Google Earth and Vaihingen datasets, is made up of ten classes that are divided into two splits: $\{\text{NWPU VHR-10}^i, i \in \{0, 1\}\}$. We adopt the widely used class settings from [46], [47], [48] to ensure a fair comparison with most existing methods. For each episode of model training, we randomly select $K \in \{3, 5, 10, 20\}$ instances of the combination of the training set and the validation set in both the benchmarks.

2) Evaluation: To thoroughly evaluate the proposed method, we follow the widely used meat-learning protocol [12], [14] on the setting of FSOD. We use K instances from both the base and novel classes, organized as a support set for the fine-tuning episode. Considering the influence of selected support images on detection performance, we report the average results of five-runs with different seeds to ensure the final result is relatively stable. As for evaluation metrics, we adopt the mean average precision (mAP) with an intersection over union (IoU) threshold at 0.5 to make a fair comparison.

B. Training Details

The training procedure for our proposed method is divided into two phases: meta-base-training and meta-fine-tuning.

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TABLE I
NOVEL SPLIT SETTINGS ON THE BOTH DIOR AND NWPU VHR-10.V2 DATASETS

Dataset	Split	Novel Classes				
DIOR	1	Baseball field	Basketball court	Bridge	Chimney	Ship
	2	Airplane	Airport	Expressway toll station	Harbor	Ground track field
	3	Dam	Golf field	Storage tank	Tennis court	Vehicle
	4	Expressway service area	Overpass	Stadium	Train station	Windmill
NWPU VHR-10.v2	1	Airplane	Baseball diamond	Tennis court	–	–
	2	Basketball court	Ground track field	Vehicle	–	–

We use the pretrained ResNet50 as a backbone network and fix the weights of three blocks in the second stage. It has been noted that an excessive number of training iterations in the fine-tuning stage may result in overfitting to the training set. Therefore, the training iterations in the base-training stage are set to 48 000 and adjusted based on the shots of support sets during the second stage. The baseline method in our research is attention-RPN [62], which incorporates a two-way contrastive training strategy and a multirelation detection head. The batch size and the learning rate are set to 8 and 8e-3, respectively. We follow the data augmentation techniques in [63] for training and resize the support images to a 320 × 320 square. To improve the computational efficiency, we principally address the K-shot setting by means of averaging activation maps. All the experiments are performed on NVIDIA TITAN XP GPUs with the PyTorch framework.

C. Comparisons to State-of-the-Arts

Tables II and III exhibit the quantitative results of the proposed GIDR on the DIOR and NWPU VHR-10.v2 benchmarks. Our method achieves superior performance in most settings on both the datasets. Specifically, on the DIOR dataset, GIDR delivers mAP scores of 33.8%, 35.9%, 40.1%, and 43.8% under different shots in the first split, exceeding the baseline methods by notable margins of 6.1%, 5.6%, 3.6%, and 4.1%, respectively. Compared with other methods following the same meta-learning paradigm, such as Meta R-CNN [40], P-CNN [10], DAnA [12], FCT [42], and MSOCL [46], GIDR yields the most substantial performance gains. While DAnA [12] and FCT [42] show strong performance in natural images, they fall short of satisfactory results in remote sensing images. This phenomenon underscores that our proposed GIDR, which establishes the correlation between query and support sets based on regional similarity, is more applicable for FSOD in remote sensing scenarios. Furthermore, the effectiveness of GIRD is also confirmed on the NWPU VHR 10.v2 dataset, where it achieves an average improvement of 3.0% in mAP over the second-best method in all the settings of the first split. We present more qualitative results on the DIOR dataset to better analyze and understand the proposed GIDR, as shown in Figs. 3 and 4. Fig. 3 illustrates the generated results of novel class proposals, and Fig. 4 presents the final classification results in both the baseline and our methods. We can easily find that our proposed GIDR generates more high-quality proposals than the baseline approach from Fig. 3. Furthermore, the detection results illustrated in Fig. 4 reveal that our method effectively reduces the false negatives in few-shot remote sensing detection and performs well even on some challenging categories, such as the ship class in the fourth column.



Fig. 3. The visualization of generated proposals under ten-shot setting in the first split on the DIOR dataset. The green and red boxes indicate the generated proposals and missing objects in the baseline method, respectively. Our proposed method exhibits better recall rates for novel classes.

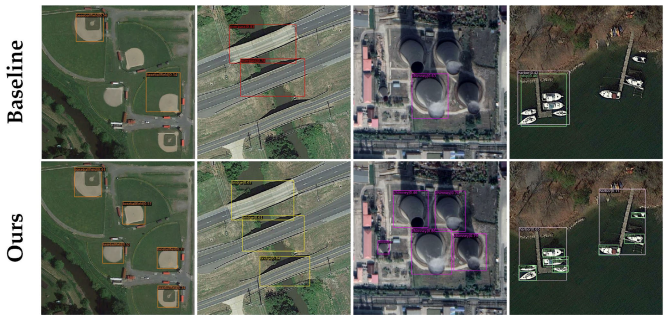


Fig. 4. Visualization of detection results under ten-shot setting in the first novel split on the DIOR dataset. It can be easily found that our proposed method has better detection accuracy performance than the baseline method.

D. Ablation Studies

We perform diverse ablation studies to validate the rationale and effectiveness of the GIDR framework in the first split on the DIOR benchmark. Attention-RPN [62] served as the baseline approach for the experiments in this section unless otherwise stated.

1) Component Analysis: Table IV shows the ablation studies on the proposed GII and DRI components. The GII module consists of two major parts: foreground enhancement (FE) and information concentration (IC). In summary, we can draw the following conclusions.

1) The GII module significantly contributes to overall improvement, achieving an average mAP result of 3.4% under different shots. One possible explanation is that

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TABLE II

PERFORMANCE COMPARISON ON DIOR USING THE MAP (%) METRIC IN THE FOUR SPLITS. * INDICATES THE METHODS THAT ARE IMPLEMENTED USING THE SAME FRAMEWORK FOR FAIR COMPARISON. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST ONES ARE UNDERLINED

Method	Split1				Split2				Split3				Split4			
	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
MetaYOLO [39]	7.5	12.1	18.1	22.0	4.8	7.0	9.0	14.1	7.8	13.7	13.8	18.5	3.7	6.8	7.2	12.2
Meta R-CNN [40]	12.0	13.0	14.1	14.4	8.8	10.8	14.9	16.7	9.1	12.2	11.9	16.1	13.9	15.8	15.0	18.1
RepMet [41]	11.3	11.3	12.1	12.2	5.2	5.6	5.9	6.8	5.5	5.7	7.4	7.8	10.9	11.6	12.8	16.7
FsDetView [36]	13.1	14.2	16.0	18.0	10.8	9.6	13.5	14.7	7.4	12.6	11.4	17.0	14.2	15.9	15.3	16.9
TFA w/cos [35]	16.0	15.3	16.4	18.9	6.8	7.5	8.9	11.0	8.7	9.3	12.1	16.9	9.5	13.8	13.8	16.6
P-CNN [10]	18.0	22.8	27.6	29.6	14.5	14.9	18.9	22.8	16.5	18.8	23.3	28.8	15.2	17.5	18.9	25.7
DAnA* [12]	25.6	28.9	32.1	34.4	12.3	14.7	17.9	20.6	15.7	16.2	21.7	25.3	14.6	16.7	19.3	23.5
FSCE [37]	27.9	28.6	33.0	37.5	13.1	14.0	15.7	20.9	15.5	16.2	23.7	28.8	17.4	20.4	22.2	24.9
MM-RCNN [11]	19.8	23.9	28.8	30.8	15.6	15.5	20.1	23.9	16.7	19.7	25.0	30.5	16.4	18.7	20.3	27.1
MSOCL [46]	24.9	27.2	33.3	39.2	13.3	13.4	15.0	18.1	13.1	15.0	23.3	27.4	10.4	12.2	16.6	22.6
FCT* [42]	16.3	21.4	27.6	33.4	10.1	12.3	15.7	19.6	14.7	16.5	21.3	26.8	11.9	13.4	19.6	22.6
TEMO [47]	28.0	31.0	35.3	40.2	15.5	17.9	21.0	23.0	16.0	22.8	24.6	30.5	17.1	21.2	26.7	32.0
Incremental-DETR [57]	27.1	30.2	36.4	37.5	12.5	15.0	20.1	20.1	16.3	22.7	25.0	31.3	14.4	20.5	25.6	28.4
G-FSDet [58]	27.5	30.5	37.4	39.8	14.1	15.8	20.7	22.6	16.0	23.2	26.2	32.0	16.7	21.0	25.8	31.7
SAE-FSDet [49]	28.8	32.4	37.0	42.4	13.9	15.6	17.4	21.3	16.7	19.0	28.4	29.8	17.2	20.4	22.6	26.7
FSIOD-DFP [50]	29.2	31.3	37.9	41.0	**17.9**	**20.1**	21.8	23.9	16.9	25.3	**29.5**	32.0	18.0	21.8	27.7	34.5
Baseline	27.7	30.3	34.5	38.7	12.8	14.8	18.9	20.7	15.7	21.3	23.8	27.6	16.9	20.3	26.4	31.9
Ours-**GIDR**	**33.8**	**35.9**	**40.1**	**43.8**	16.2	18.7	**22.1**	**24.4**	**19.8**	**25.4**	28.7	**32.4**	**21.9**	**25.1**	**32.2**	**37.7**

TABLE III

PERFORMANCE COMPARISON ON NWPU VHR-10.V2 USING THE MAP (%) METRIC IN THE TWO SPLITS. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST ONES ARE UNDERLINED

Method	Publication	Split1				Split2			
		3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
MetaYOLO [39]	ICCV'19	15.3	16.2	24.0	27.1	28.6	31.4	32.5	33.9
Meta R-CNN [40]	ICCV'19	20.5	21.7	26.9	28.2	21.4	35.3	37.1	39.4
FsDetView [36]	ECCV'20	24.5	29.5	31.7	32.7	39.0	40.3	45.0	46.2
TFA w/cos [35]	ICML'20	8.8	9.4	9.2	10.8	11.1	12.4	11.3	11.5
P-CNN [10]	TGRS'21	41.8	49.1	63.2	66.8	39.3	46.1	55.9	58.3
DeFRCN [38]	ICCV'21	37.9	46.0	62.9	64.6	39.1	45.5	54.0	57.3
iMTFA [59]	CVPR'21	43.2	49.9	70.0	72.7	46.2	54.3	62.5	66.3
MSOCL [46]	TGRS'22	63.5	59.3	69.9	84.2	48.0	48.7	53.2	74.1
TEMO [47]	TGRS'23	57.5	66.7	75.0	75.8	50.8	**59.3**	64.5	73.3
Incremental-DETR [57]	AAAI'23	50.9	53.1	69.9	73.1	47.3	57.2	65.6	70.5
G-FSDet [58]	ISRPS'23	49.0	56.1	71.8	75.4	50.0	58.7	67.0	75.8
SAE-FSDet [49]	TGRS'24	57.9	59.4	71.0	85.0	–	–	–	–
Baseline	–	62.7	68.9	73.6	81.0	47.0	54.3	63.7	72.1
Ours-**GIDR**	–	**67.3**	**73.0**	**78.1**	**86.0**	**50.9**	58.5	**68.3**	**76.4**

the absence of foreground proposals has a more significant impact than the mismatch of two branches within the meta-learning framework.

2) Building on the FE strategy, the IC component (scheme (c)) leads to an average improvement of 1.3%. This outcome suggests that the global support information plays a meaningful role in localizing the foreground targets of novel classes.
3) To validate the superiority of the DRI module, we compared it with the augmentation method of random rotation [64] [scheme (d)] in Table IV. The results indicate that the augmentation technique of random rotation enhances detection accuracy by approximately 1% when the support set contains more samples. However, its efficacy is diminished when the number of instances is limited. As a comparison, the DRI module exhibits substantial improvements on all the settings. We argue that even with multiangle rotations of training samples, the random rotation strategy struggles to achieve complete alignment of the targets. The primary innovation of our proposed DRI module is its independence from direction alignment between support and query samples. The utilization of offset learning facilitates the adaptive establishment of positional correlations between the targets in the two branches.
4) These components can be considered complementary (scheme (e)), collectively contributing to the final detection results of 6.1%. Intuitively, it is clear that as the

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TABLE IV

ABLATION STUDIES ON GII AND DRI IN MAP METRIC. FE AND IC INDICATE THE FOREGROUND ENHANCEMENT AND THE INFORMATION CONCENTRATION PROPOSED IN SECTION III-C, RESPECTIVELY. RR INDICATES THE AUGMENTATION METHOD OF RANDOM ROTATION [64]

Scheme	GII		RR	DRI	3-shot	5-shot	10-shot	20-shot
	FE	IC						
(a)					27.7	30.3	34.5	38.7
(b)	✔				30.2	32.7	36.4	40.5
(c)	✔	✔			31.6	33.8	37.8	41.7
(d)			✔		28.1	30.7	35.7	39.6
(e)				✔	29.9	32.4	36.8	40.9
(f)	✔	✔		✔	**33.8**	**35.9**	**40.1**	**43.8**

TABLE V

ABLATION STUDIES OF THE DIFFERENT SIZES OF KERNELS IN GII

Scheme	1×1	3×3	7×7	3-shot	5-shot	10-shot	20-shot
(a)	✔			32.0	34.0	38.5	42.1
(b)		✔		32.9	35.1	39.2	42.9
(c)			✔	32.3	34.2	38.7	42.4
(d)	✔	✔		33.3	35.5	39.6	43.3
(e)	✔	✔	✔	**33.8**	**35.9**	**40.1**	**43.8**

TABLE VI

ABLATION STUDIES ON DIFFERENT KERNELS IN GII FOR TARGETS OF DIFFERENT SIZES. $\mathbf{AP_S}$, $\mathbf{AP_M}$, AND $\mathbf{AP_L}$ INDICATE THE AP VALUE FOR SMALL, MEDIUM, AND LARGE TARGET OBJECTS, RESPECTIVELY

Kernel Scale	1×1	3×3	7×7	AP_S	AP_M	AP_L
(a)				4.8	34.4	64.7
(b)	✔			5.6	35.2	65.6
(d)	✔	✔		6.1	37.6	67.1
(e)	✔	✔	✔	**6.1**	**38.3**	**69.3**

TABLE VII

ABLATION STUDIES ON THE DIFFERENT INTERACTIVE STRATEGIES BETWEEN THE QUERY AND SUPPORT SETS FOR PROPOSAL GENERATION

Strategy	3-shot	5-shot	10-shot	20-shot
Attention-RPN [62]	27.7	30.3	34.5	38.7
Dense Relation [65]	26.6	29.4	32.3	36.2
Spatial Attention [66]	25.1	28.0	31.4	34.9
DAnA [12]	28.5	31.6	33.9	37.4
Ours-**GII**	**31.6**	**33.8**	**37.8**	**41.7**

number of high-quality proposals increases, the impact of the proposed DRI becomes more pronounced.

2) Kernel Scales in GII: We conduct a series of ablation studies to investigate the impact of different kernel sizes on GII, as depicted in Tables V and VI. The experiments in this section are based on scheme (e) as shown in Table IV, where we remove kernels of different sizes to evaluate their effectiveness individually. Our initial findings indicate that the 1×1 kernel yields less contribution to the GII module than the 3×3 kernels in Table V. The results suggest that retaining certain spatial information from the support sets is essential for the performance of our proposed method. Specifically, we observed that the 1×1 kernel significantly enhances the detection of small targets, whereas the 7×7 kernel demonstrates superior performance in detecting large targets in Table VI. Furthermore, the 3×3 kernel consistently improves detection accuracy across all the target sizes, offering a well-balanced contribution. These results emphasize that the proposed support kernels of varying sizes are highly effective in addressing the challenges of multiscale target detection in remote sensing images.

3) Query–Support Interactive Strategy: Table VII presents the ablation studies for the different interactive strategies between query and support sets. It can be easily observed that our proposed GII achieves a notable improvement in the FSOD models compared with other competitors. Furthermore, the foreground activation maps are visualized for a more detailed comparison with different interactive strategies, as illustrated in Fig. 5. Compared with the Attention-RPN [62] and DAnA [12] methods, the foreground activation maps generated from our proposed GII concentrate more on query targets and have less interference, particularly for the categories which lack fixed shapes and are challenging to distinguish from background (e.g., expressway service areas and golf courses). We attribute this phenomenon to the fundamental differences in the interaction patterns between query and support features. Attention-RPN [62] used class prototypes to adjust the weights of query feature channels, which overlook the spatial information in the support sets. DAnA [12] designed pixel-level relationships between the features of two branches that inevitably introduce additional background noise. The core of GII lies in regional similarity, which more accurately captures the correlation between the query and support objects in remote sensing images. In addition, Table VIII presents the detection accuracy for all the categories on ten-shot setting in the first split. The results demonstrate that our approach achieves significant improvements for both the base and novel classes. Moreover, it showcases strong capabilities in accurately localizing foreground objects within complex background and effectively matching targets of two branches with diverse orientations, such as airplanes and ships.

4) Number of Channel Groups in DRI: We investigate the impact of split channel groups N_r in Section III-D, as summarized in Table IX. Analysis reveals that the detection accuracy shows a pattern of early increase and later decline, with the best results obtained when setting N_r to 8. We attribute the early improvement in detection accuracy to the effective utilization of support information across various channels. One potential reason for the subsequent decline is that the drastic decrease in parameters of our proposed DRI prevents the complete learning of kernels and their corresponding offsets.

5) Visualization of Aligned Activation Maps and Offsets: To better analyze and understand the proposed DRI, we visualize the aligned activation maps of query RoI features, as shown in Fig. 6. Initially, the activation maps corresponding to different

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TABLE VIII
FEW-SHOT DETECTION RESULTS (%) FOR BOTH BASE AND NOVEL CLASSES ON THE DIOR DATASET IN THE FIRST SPLIT

Method	BF	BC	BR	CH	SH	APL	APO	ETS	HA	GTF	DAM	GF	STO	TC	VE	ESA	OP	STA	TS	WM
Baseline	47.2	45.6	11.4	62.1	6.2	44.1	61.5	39.5	45.5	58.1	40.2	62.4	29.5	71.1	24.2	45.1	35.6	55.5	41.9	61.7
Ours	**53.2**	**57.6**	**13.3**	**67.4**	**9.1**	**52.0**	**65.5**	**44.7**	**48.5**	**59.9**	**41.6**	**65.0**	**34.5**	**78.0**	**26.2**	**46.0**	**36.4**	**66.5**	**45.4**	**67.5**

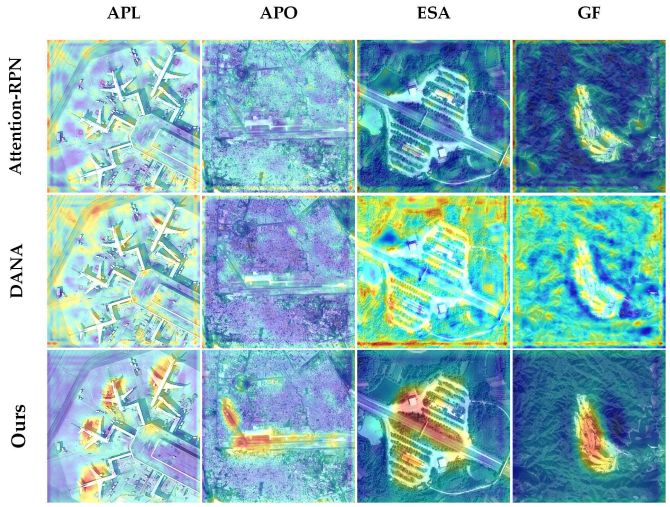


Fig. 5. Visualization of foreground activation map $\mathbf{A}_{\text{fore}}$ in different novel classes. The abbreviations APL, APO, ETS, and HBO indicate airplanes, airports, expressway toll stations, and harbors, respectively.

TABLE IX
ABLATION STUDIES ON THE NUMBER OF SPLIT GROUPS N_r IN DRI

Groups	3-shot	5-shot	10-shot	20-shot
4	33.1	35.2	39.5	43.2
8	**33.8**	**35.9**	**40.1**	**43.8**
16	33.5	35.6	39.7	43.5
32	32.6	34.7	38.8	42.7

support kernels are derived using (12). Subsequently, these maps are upsampled and normalized, and then combined with the relevant support features to generate heatmaps. In the final step, ground-truth boxes are used to create the corresponding masks, which highlight the target objects of interest. The activation regions produced by different support kernels are distinct. Taking the ship class as an example, the main activation regions of $\mathbf{M}_1$, $\mathbf{M}_3$, and $\mathbf{M}_5$ correspond to the bow, the main body, and the stern, respectively. This phenomenon demonstrates that DRI indeed aligns the relevant semantic components between the query and support objects at a fine-grained level. Moreover, the visualizations of offsets learned by our proposed DRI module are presented in Fig. 7. When the green sampling point is positioned on the foreground, the red points are largely concentrated in the corresponding target regions, such as the tail of the airplane and the hull of the ship. In stark contrast, sampling points are generally sparsely distributed in irrelevant background regions. From these observations, we argue that the designed DRI module enables the receptive fields of support kernels to adaptively

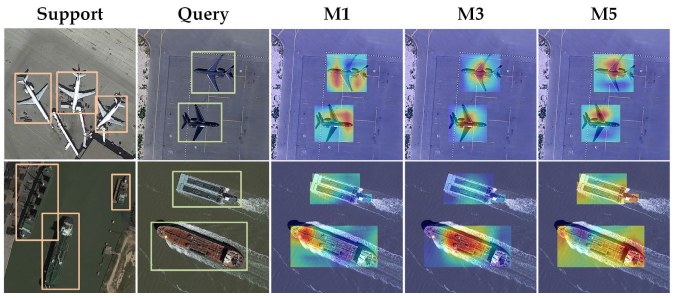


Fig. 6. Visualization of aligned activation maps $\mathbf{M}^i_{\text{align}}$ in the airplane and ship classes, where i indicates the order of activation map produced in the DRI module.

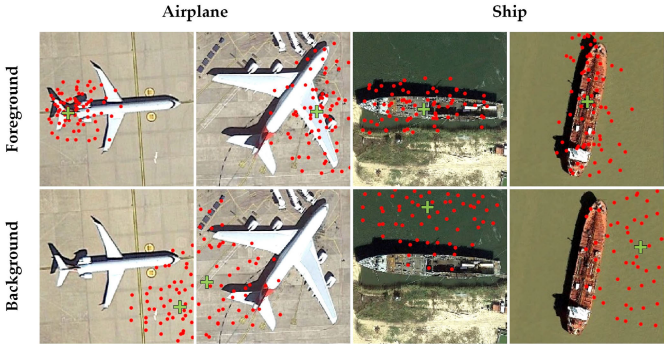


Fig. 7. Visualization of offsets learned by the designed DRI module. The first and second rows of images show the sampling locations for two different activation units (green points) on the foreground and background, respectively.

focus on the key regions of query set targets through offset learning.

V. CONCLUSION

This study proposes an innovative approach called GIDR, aimed at addressing the difficulties of foreground localization and category matching in few-shot remote sensing object detection. The core idea is based on imprinting, which uses the tailored support information to establish the regional correlation between the query and support objects from a global to local perspective. In this article, we design two simple but effective modules at different stages. In particular, the lightweight GII module enhances the quality of generated proposals by increasing the activation on foreground features in the RPN, and the well-crafted DRI module improves the classification performance by reducing the directional disparities between the query and support sets. In contrast to previous FSOD methods that tackle similar issues, our approach introduces less background noise and effectively mitigates the overfitting to training samples in FSOD models. Comprehensive experiments on standard benchmarks validate the effectiveness of each designed component. Furthermore,

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the comparison to state-of-the-art approaches underscores the advantages of our proposed GIDR and indicates potential directions for future research in network design.

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